

GGL90 Validation Report

MITgcm vs Python Port Comparison (single-step replay)

Generated: 2026-09-27 05:56:07

MITgcm Reference:

File: mitgcm_ggl90_outputs_vermix_20_1dmix024.nc
UUID: 84f5dfa8-4a53-4dae-9541-30dec502d26e

Python Port Output:

File: python_ggl90_outputs_vermix_20_1dmix024.nc
UUID: 84f5dfa8-4a53-4dae-9541-30dec502d26e

Provenance: ☐ UUIDs match

Timesteps: 20

Grid: 1 × 1 × 26

Fields compared: visc_az, diff_kz, mixing_length, tke_after
(GGL90 single-step replay: Python fed MITgcm's own captured
tke_before(t); only the one-step update is compared, isolating
agreement on the physics formula from trajectory drift.)

GGL90 Field Statistics

Wet-cell comparison (NaN-excluded below-seafloor/above-ice-shelf cells)

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visc_az  (Vertical eddy viscosity (K_m), m^2/s)
          Status: □ EXCELLENT
Median |diff|: 0.0000e+00  P95 |diff|: 0.0000e+00  Max |diff|: 1.3202e-05
Median rel:   0.0000e+00  P95 rel:   0.0000e+00  Max rel:   7.3390e-04
          Fraction >1% rel error: 0.0000%
          Points compared: 520

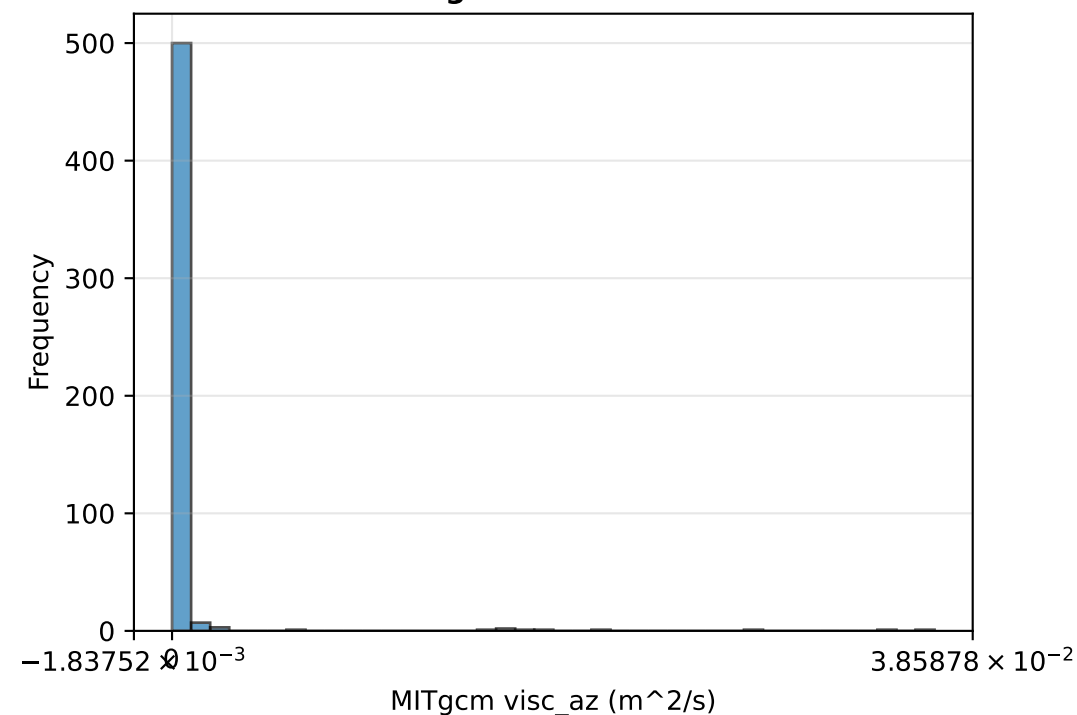
diff_kz  (Vertical eddy diffusivity (K_h), m^2/s)
          Status: □ EXCELLENT
Median |diff|: 0.0000e+00  P95 |diff|: 0.0000e+00  Max |diff|: 1.3202e-05
Median rel:   0.0000e+00  P95 rel:   0.0000e+00  Max rel:   2.1917e-03
          Fraction >1% rel error: 0.0000%
          Points compared: 520

mixing_length (GGL90 mixing length (L), m)
          Status: □ EXCELLENT
Median |diff|: 7.9744e-05  P95 |diff|: 3.3584e-04  Max |diff|: 7.7561e-03
Median rel:   6.9373e-04  P95 rel:   1.9251e-03  Max rel:   2.4250e-03
          Fraction >1% rel error: 0.0000%
          Points compared: 520

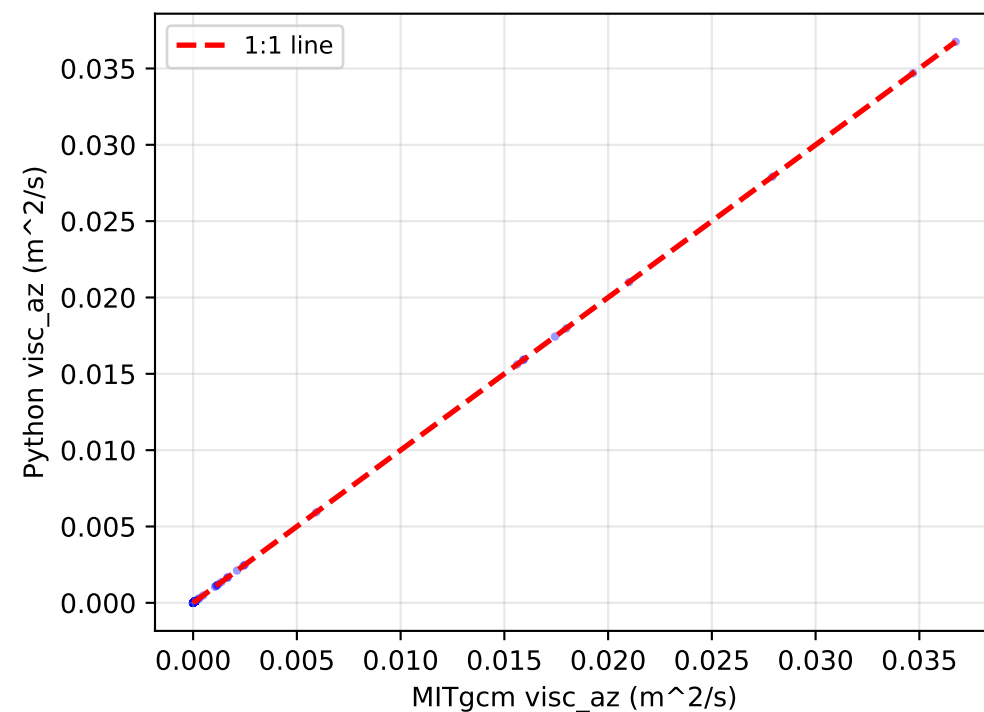
tke_after (TKE at t+1 (single-step replay), m^2/s^2)
          Status: □ EXCELLENT
Median |diff|: 0.0000e+00  P95 |diff|: 1.0188e-09  Max |diff|: 7.8363e-07
Median rel:   0.0000e+00  P95 rel:   3.0206e-04  Max rel:   3.0660e-03
          Fraction >1% rel error: 0.0000%
          Points compared: 520
```

visc_az (Vertical eddy viscosity (K_m)) -- MITgcm vs Python

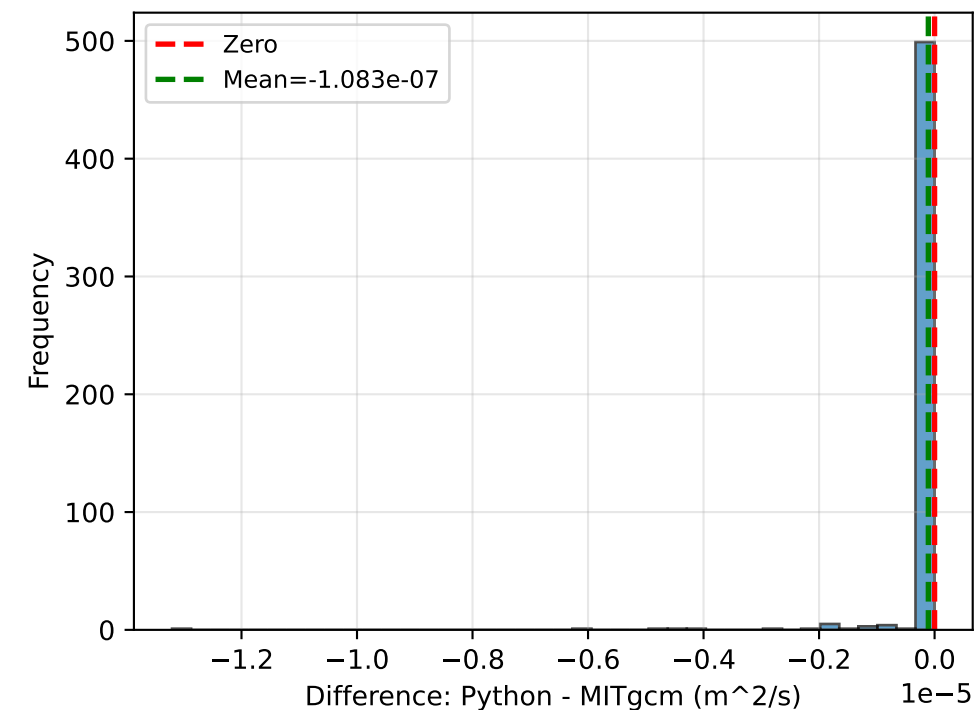
MITgcm Distribution



Scatter Plot



Difference Distribution



visc_az Statistics:

Median |diff|: 0.0000e+00
P95 |diff|: 0.0000e+00
Max |diff|: 1.3202e-05

Median rel: 0.0000e+00
P95 rel: 0.0000e+00
Max rel: 7.3390e-04

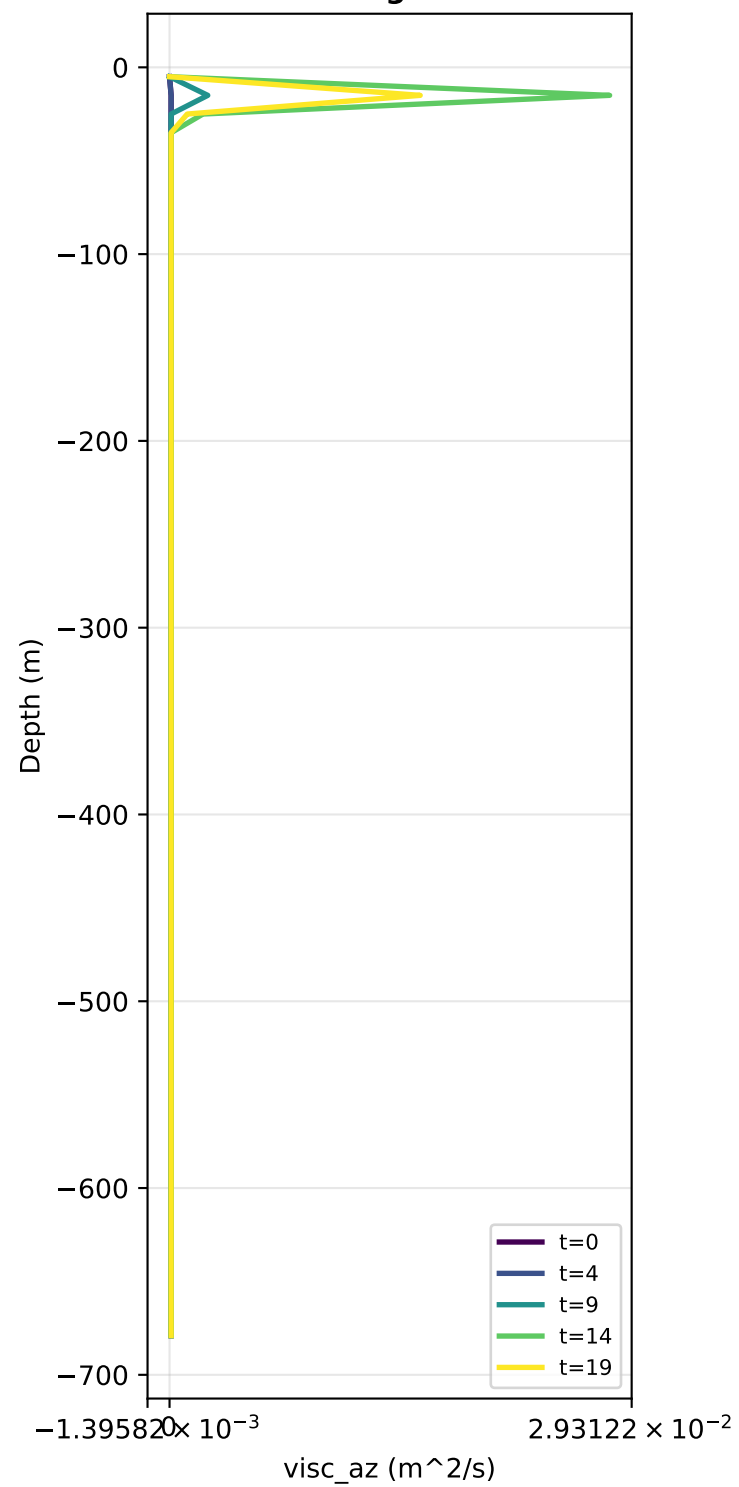
Fraction >1% rel: 0.0000%
Points compared: 520

Assessment:

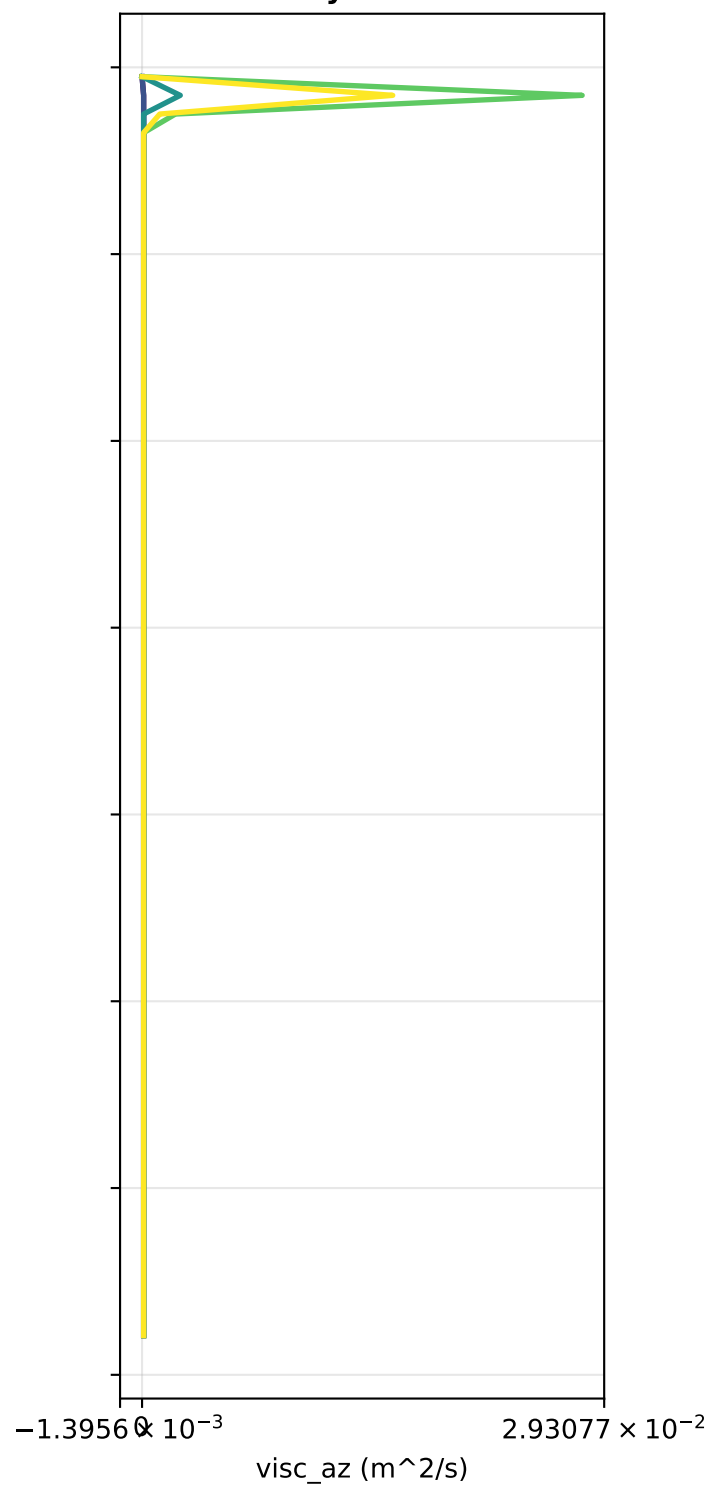
□ EXCELLENT

visc_az Vertical Profiles (Vertical eddy viscosity (K_m))

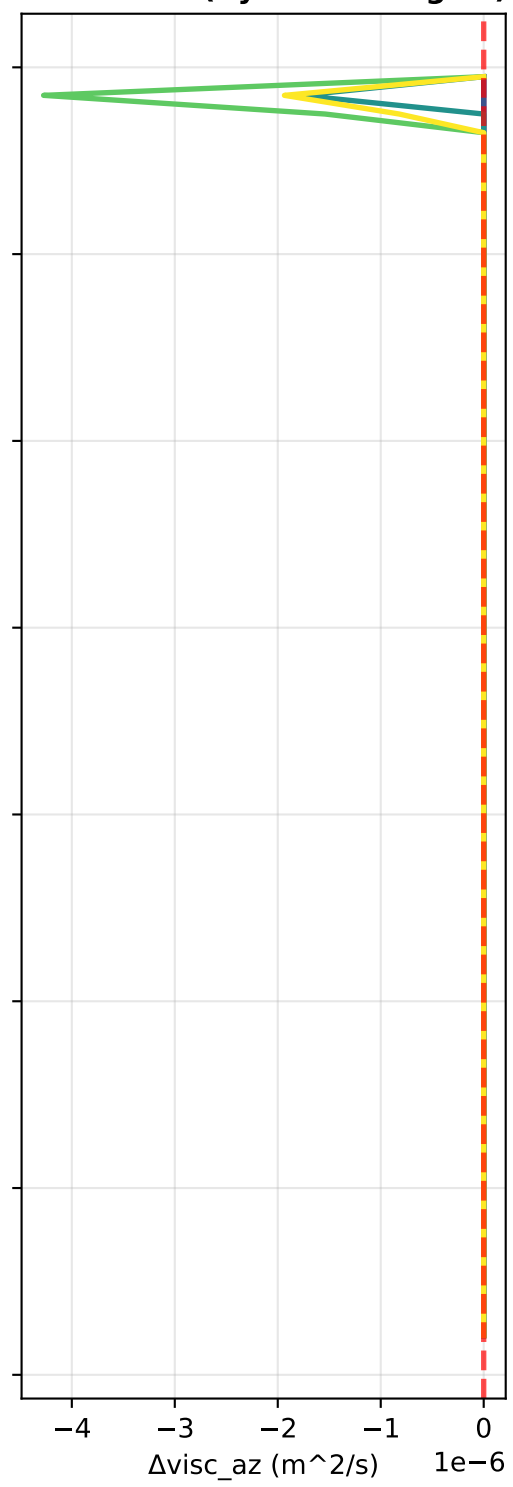
MITgcm



Python

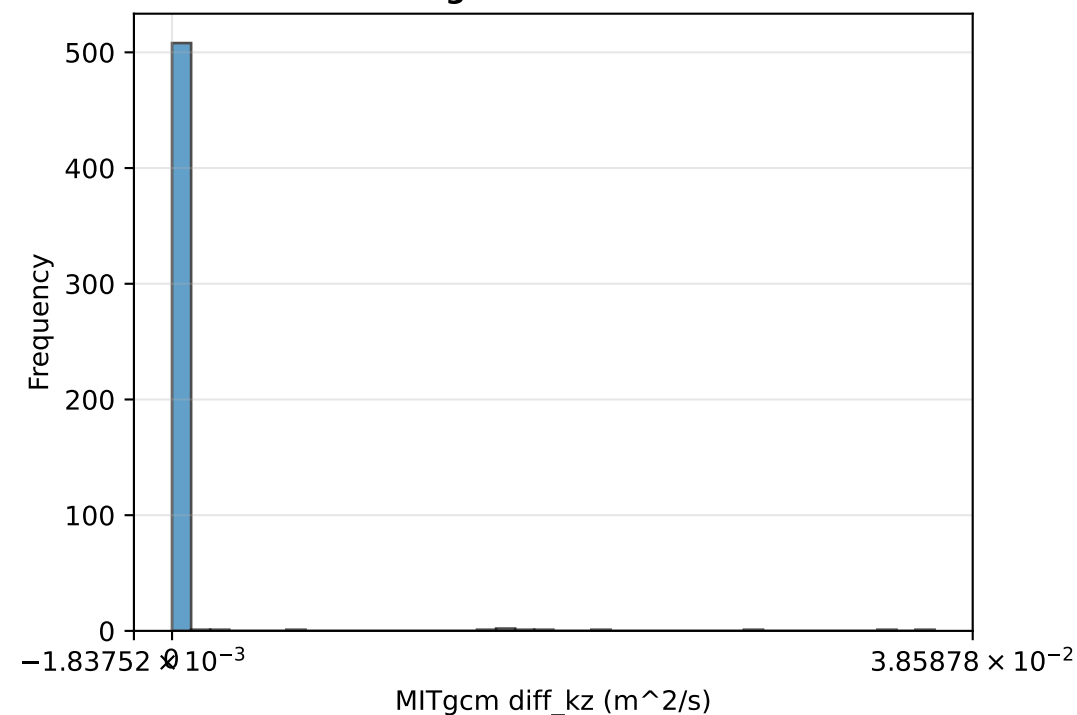


Difference (Python - MITgcm)

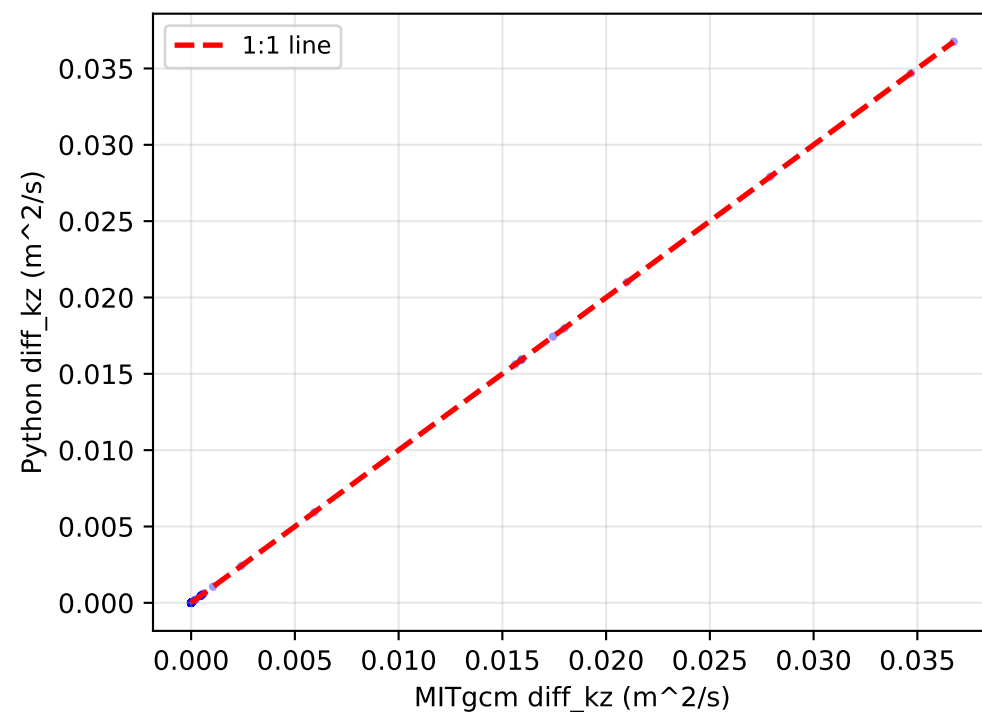


diff_kz (Vertical eddy diffusivity (K_h)) -- MITgcm vs Python

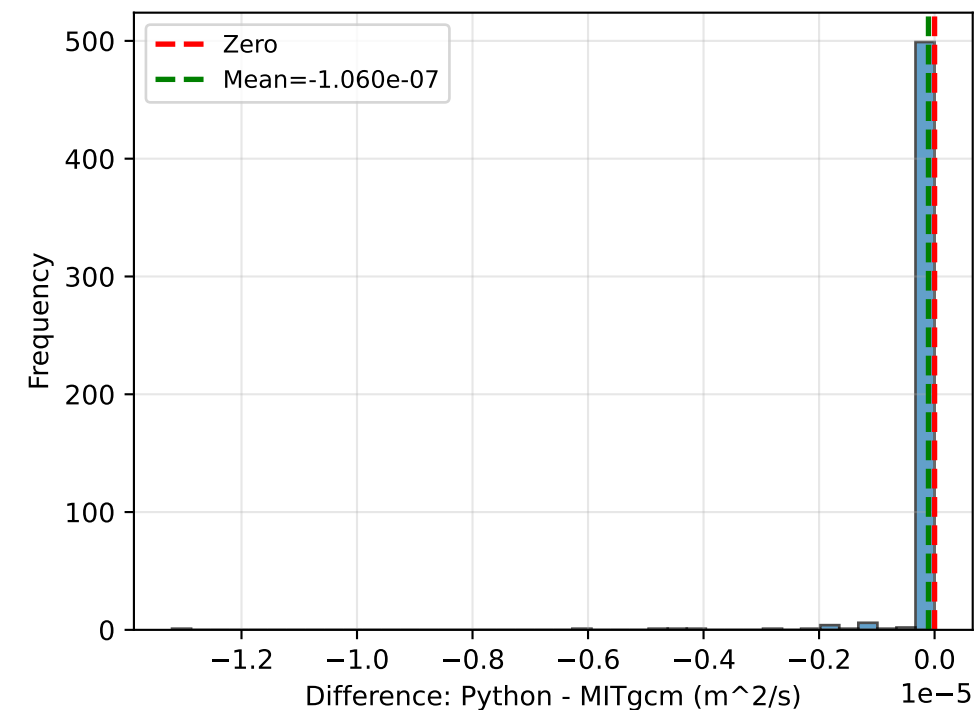
MITgcm Distribution



Scatter Plot



Difference Distribution



diff_kz Statistics:

Median |diff|: 0.0000e+00
P95 |diff|: 0.0000e+00
Max |diff|: 1.3202e-05

Median rel: 0.0000e+00
P95 rel: 0.0000e+00
Max rel: 2.1917e-03

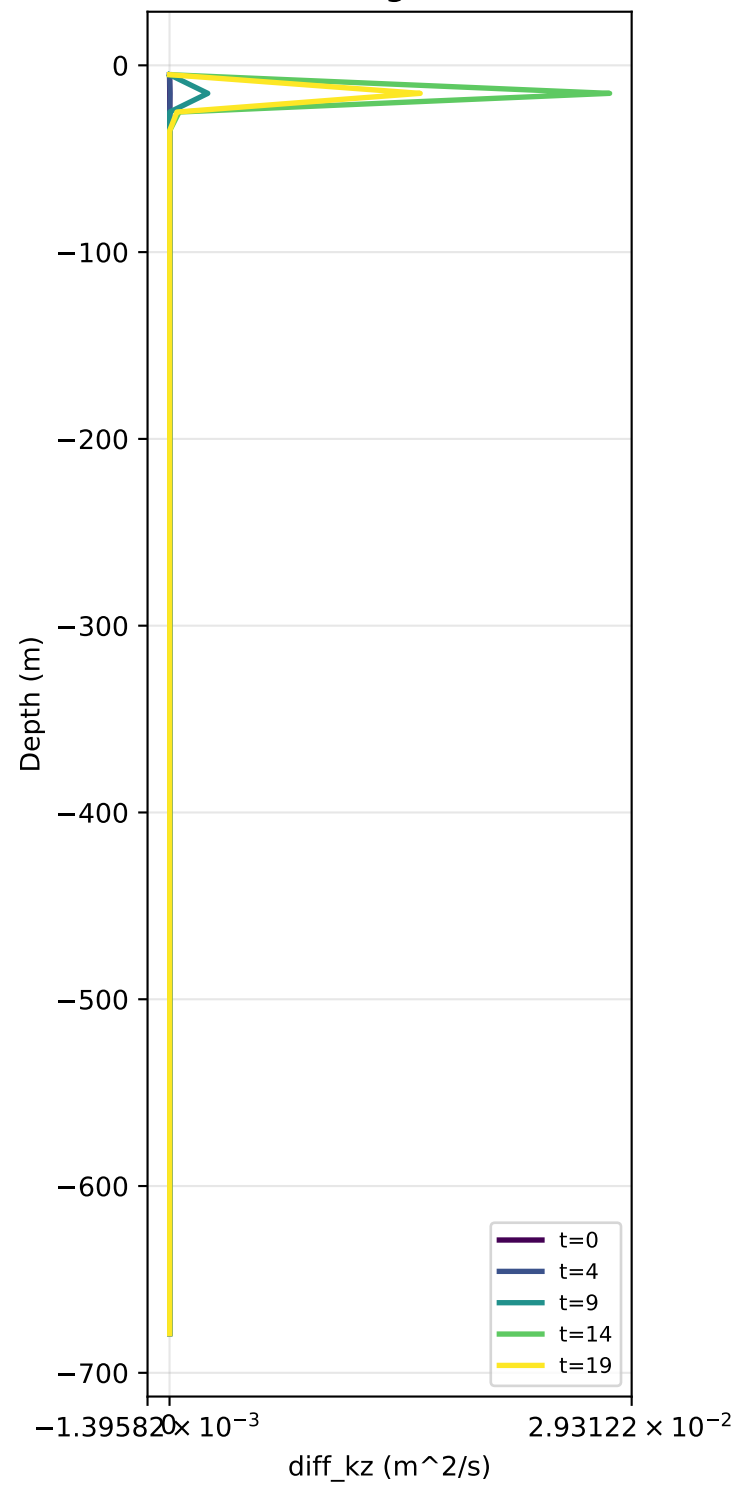
Fraction >1% rel: 0.0000%
Points compared: 520

Assessment:

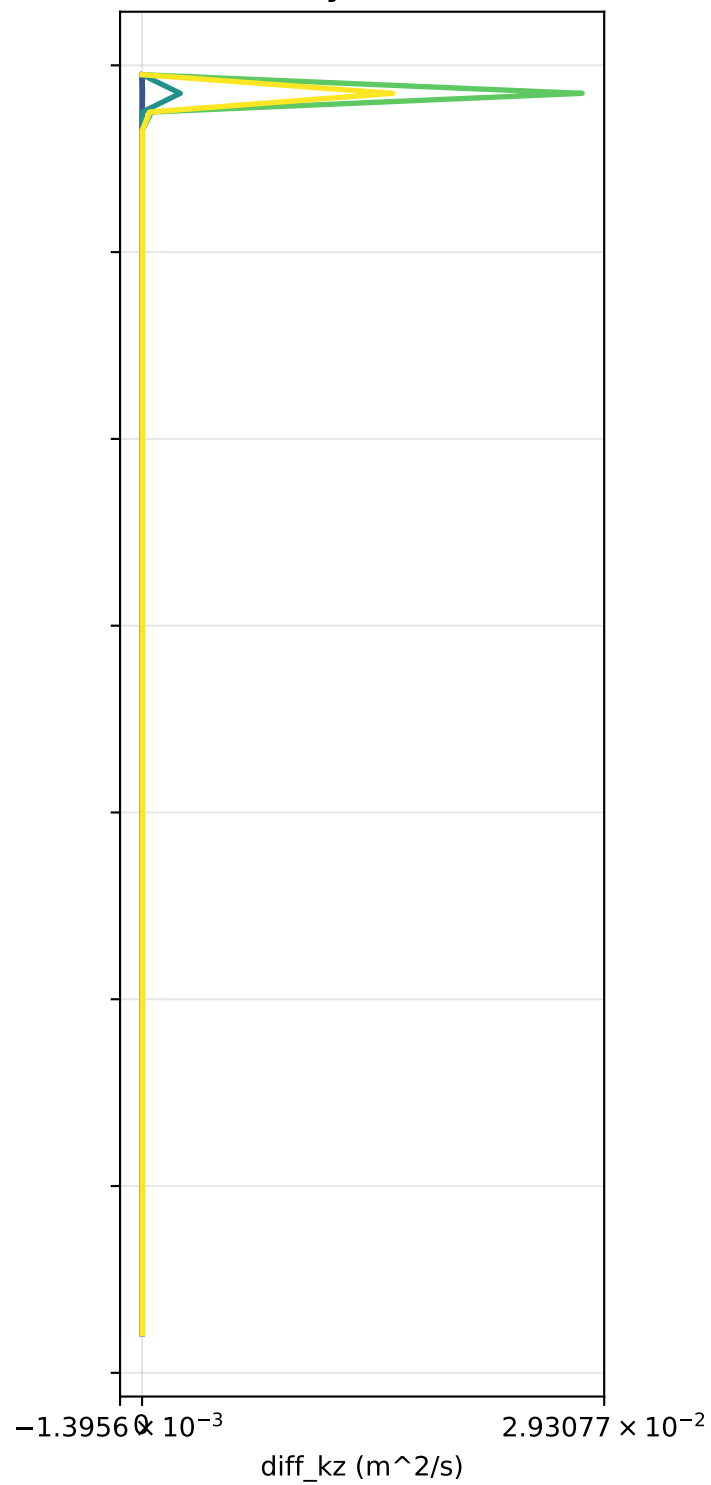
EXCELLENT

diff_kz Vertical Profiles (Vertical eddy diffusivity (K_h))

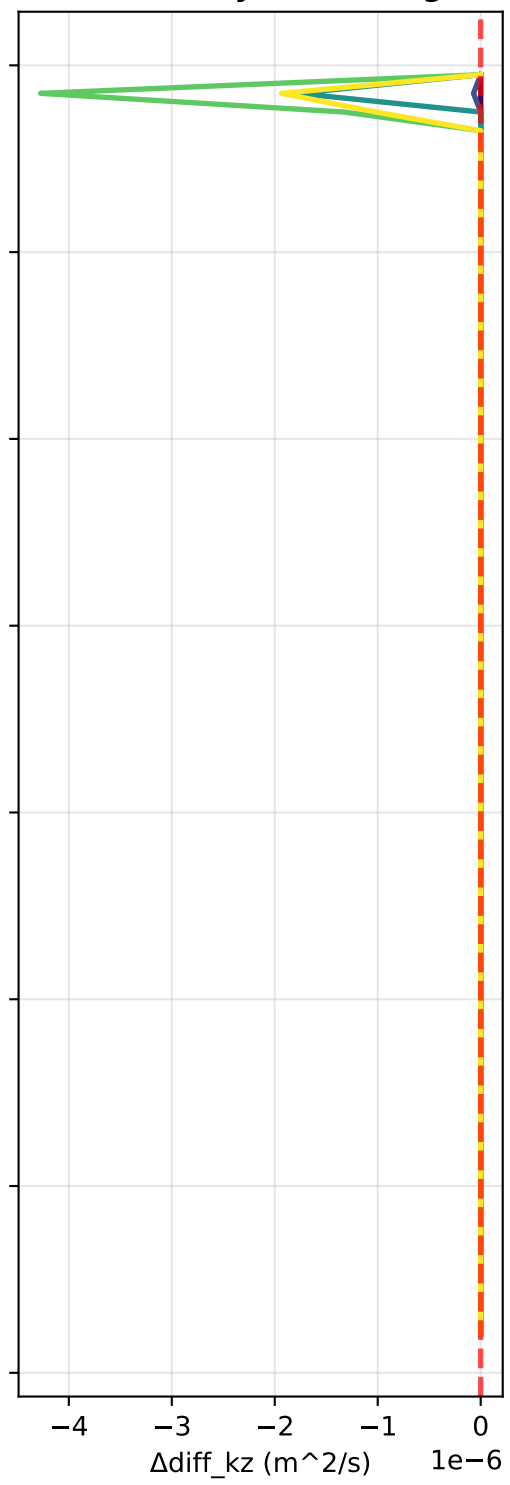
MITgcm



Python

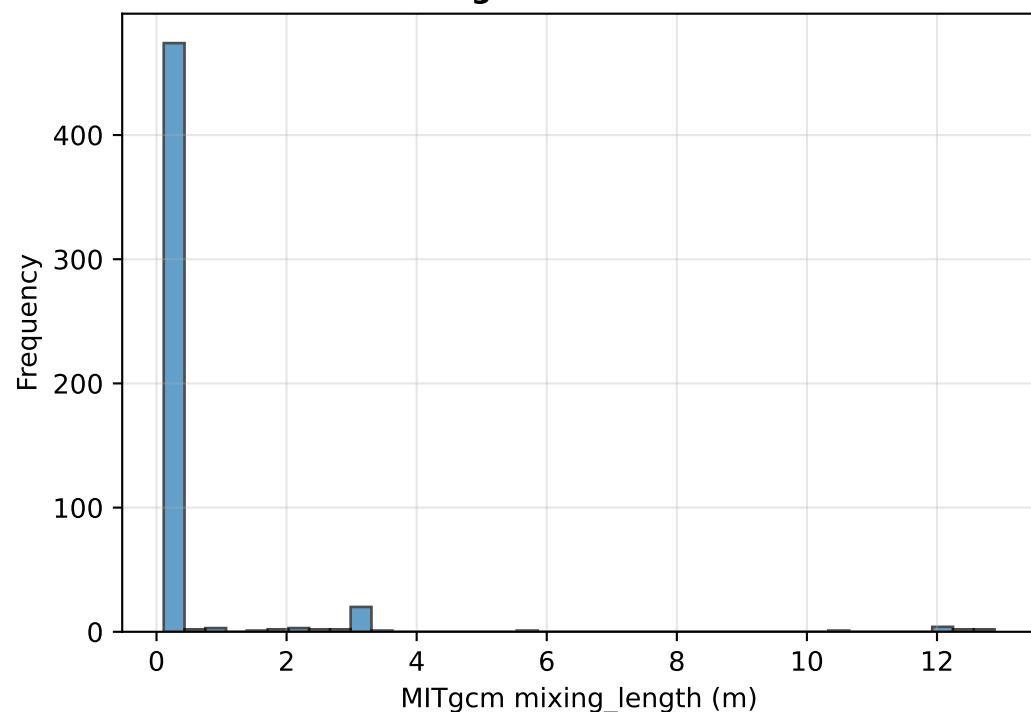


Difference (Python - MITgcm)

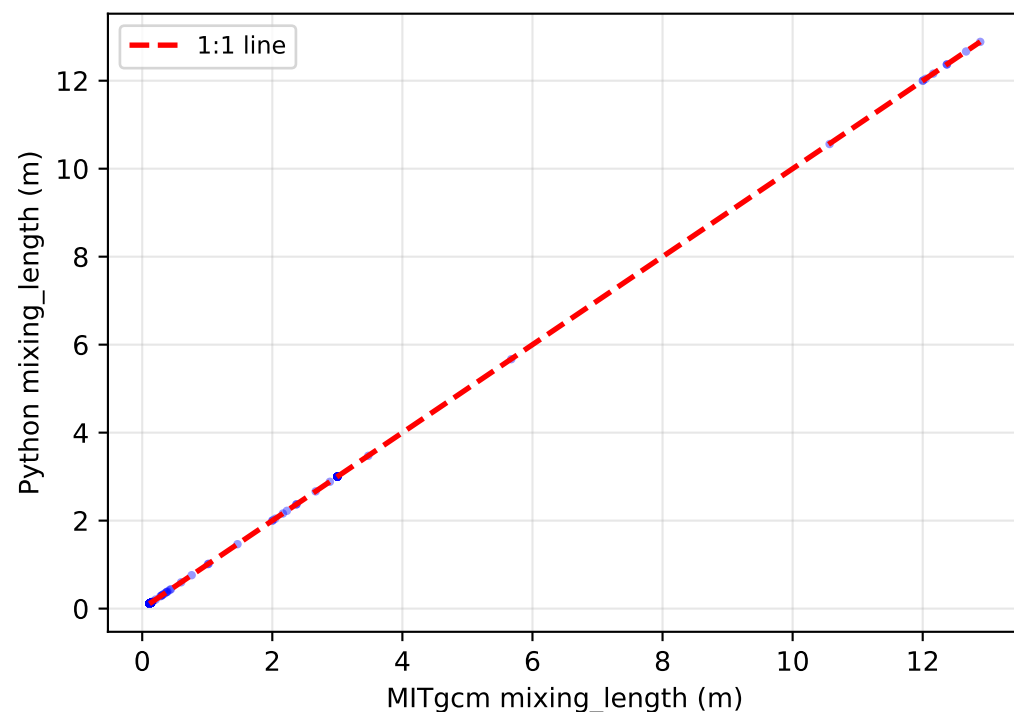


mixing_length (GGL90 mixing length (L)) -- MITgcm vs Python

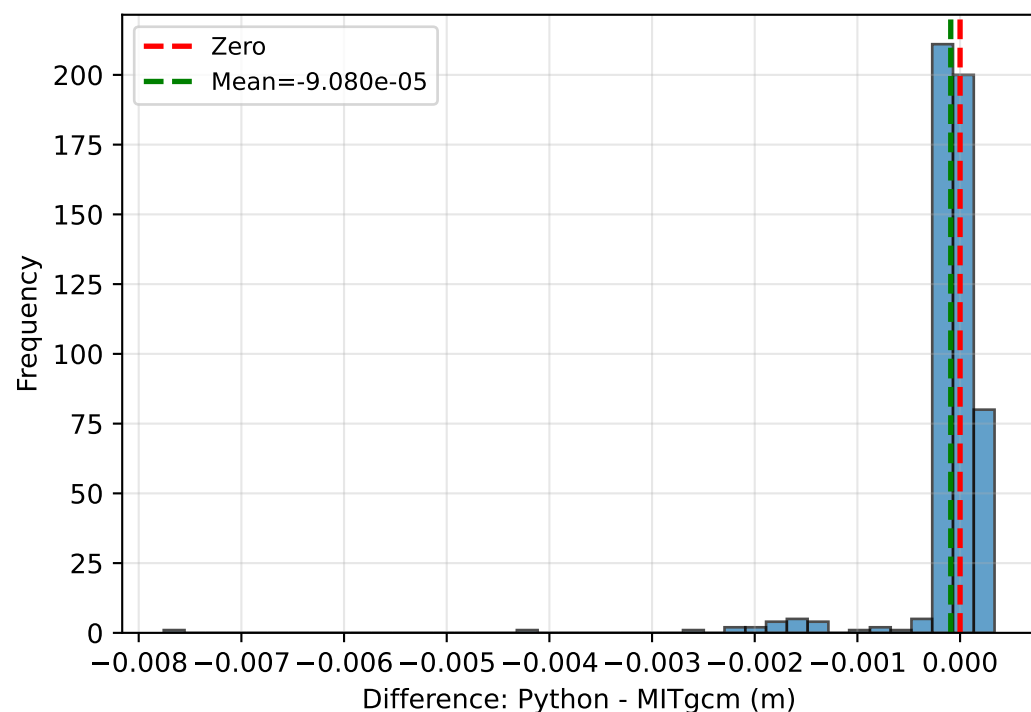
MITgcm Distribution



Scatter Plot



Difference Distribution



mixing_length Statistics:

Median |diff|: 7.9744e-05
P95 |diff|: 3.3584e-04
Max |diff|: 7.7561e-03

Median rel: 6.9373e-04
P95 rel: 1.9251e-03
Max rel: 2.4250e-03

Fraction >1% rel: 0.0000%
Points compared: 520

Assessment:

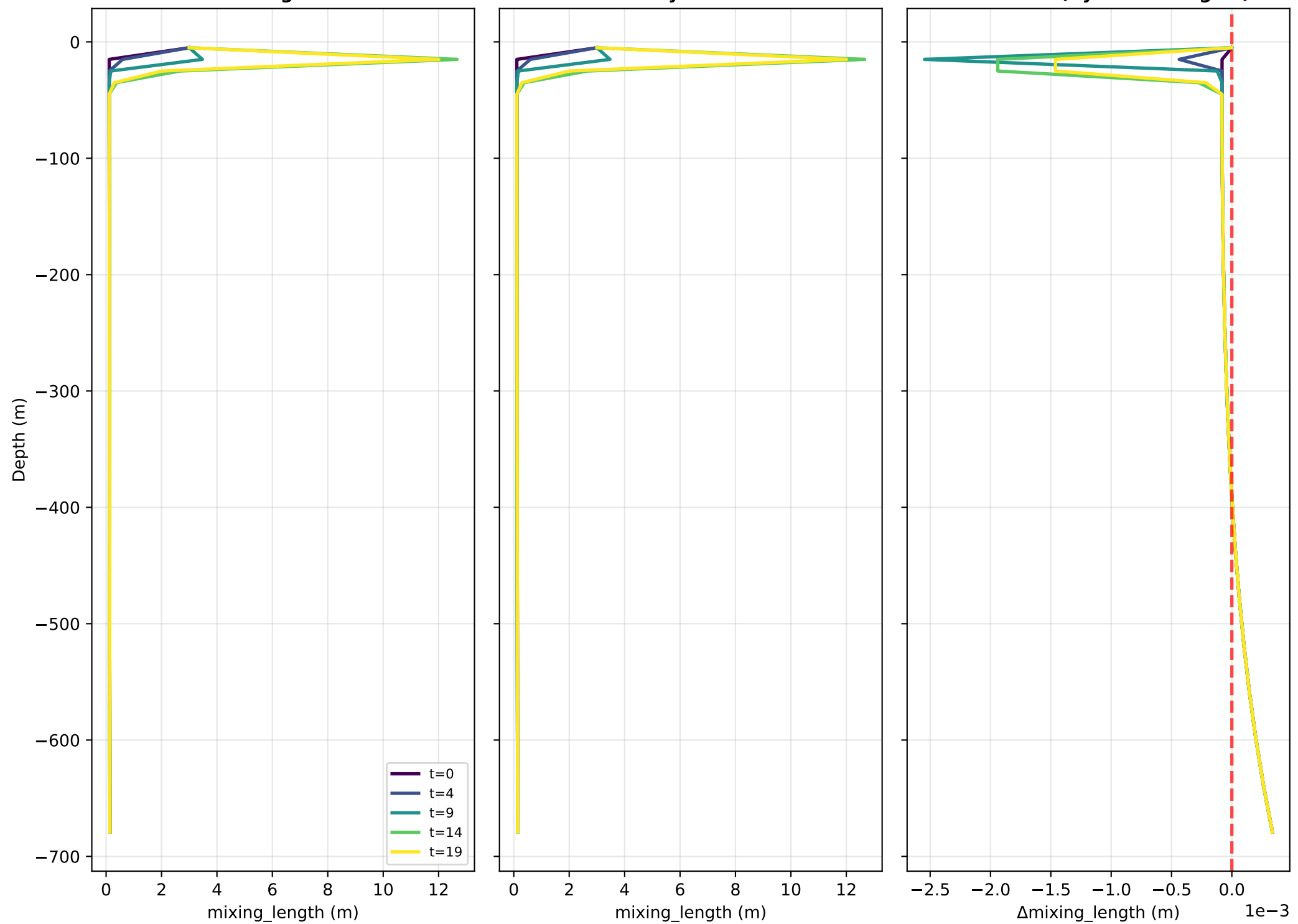
EXCELLENT

mixing_length Vertical Profiles (GGL90 mixing length (L))

MITgcm

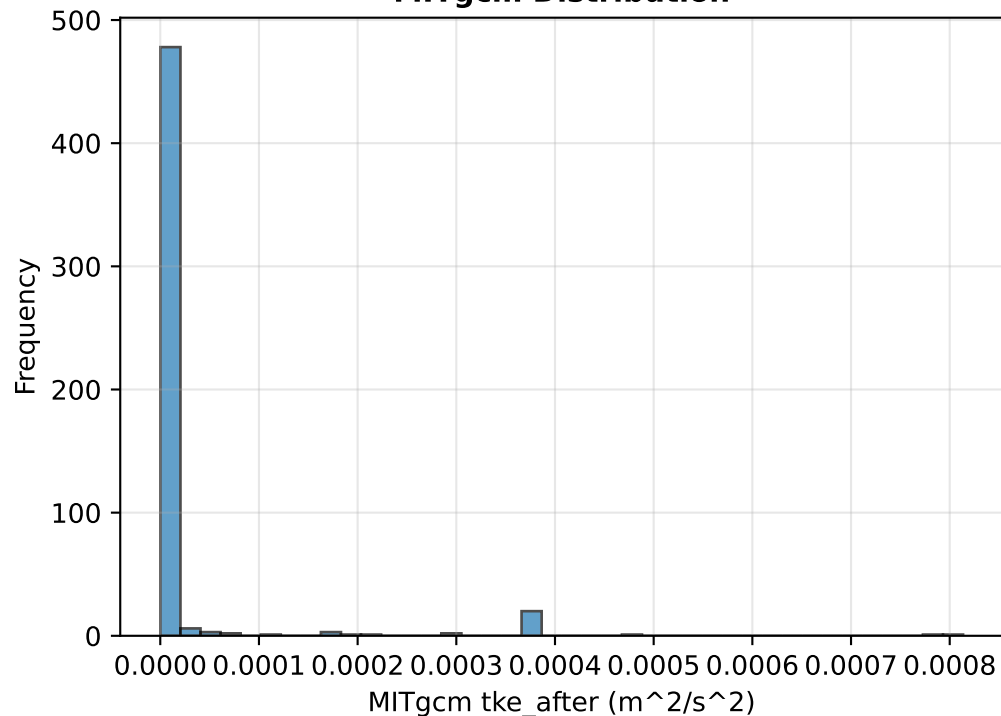
Python

Difference (Python - MITgcm)

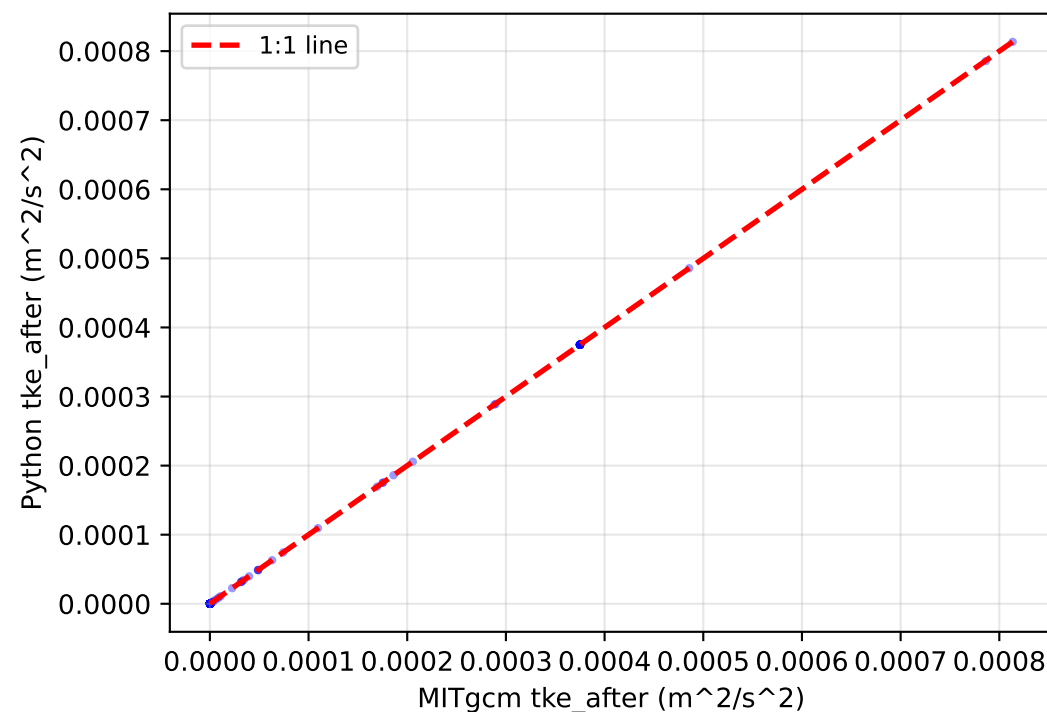


tke_after (TKE at t+1 (single-step replay)) -- MITgcm vs Python

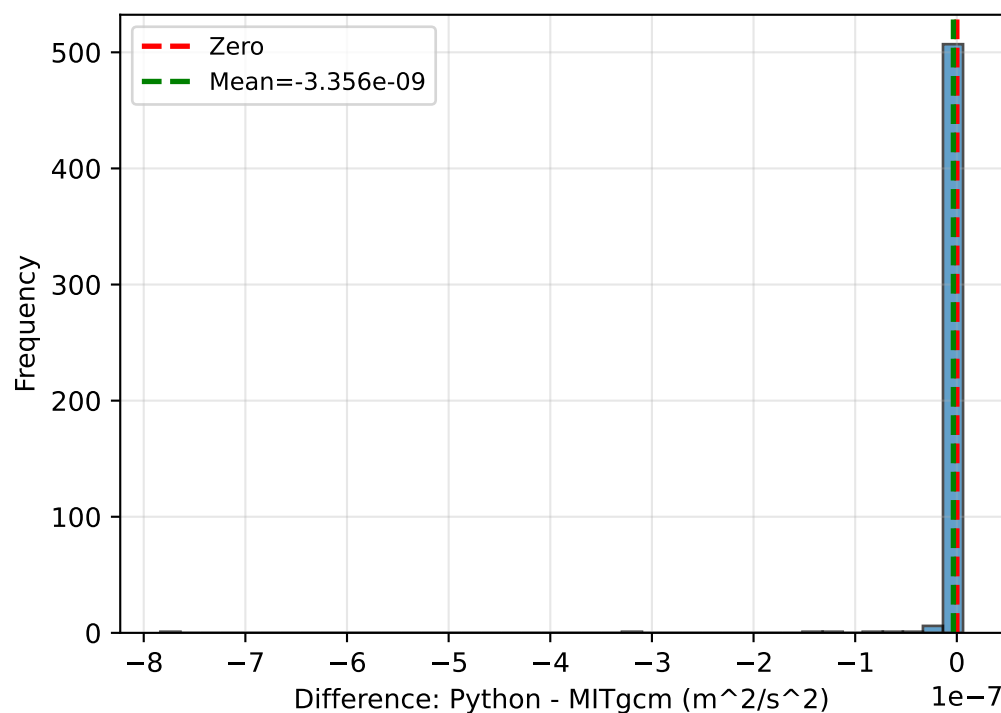
MITgcm Distribution



Scatter Plot



Difference Distribution



tke_after Statistics:

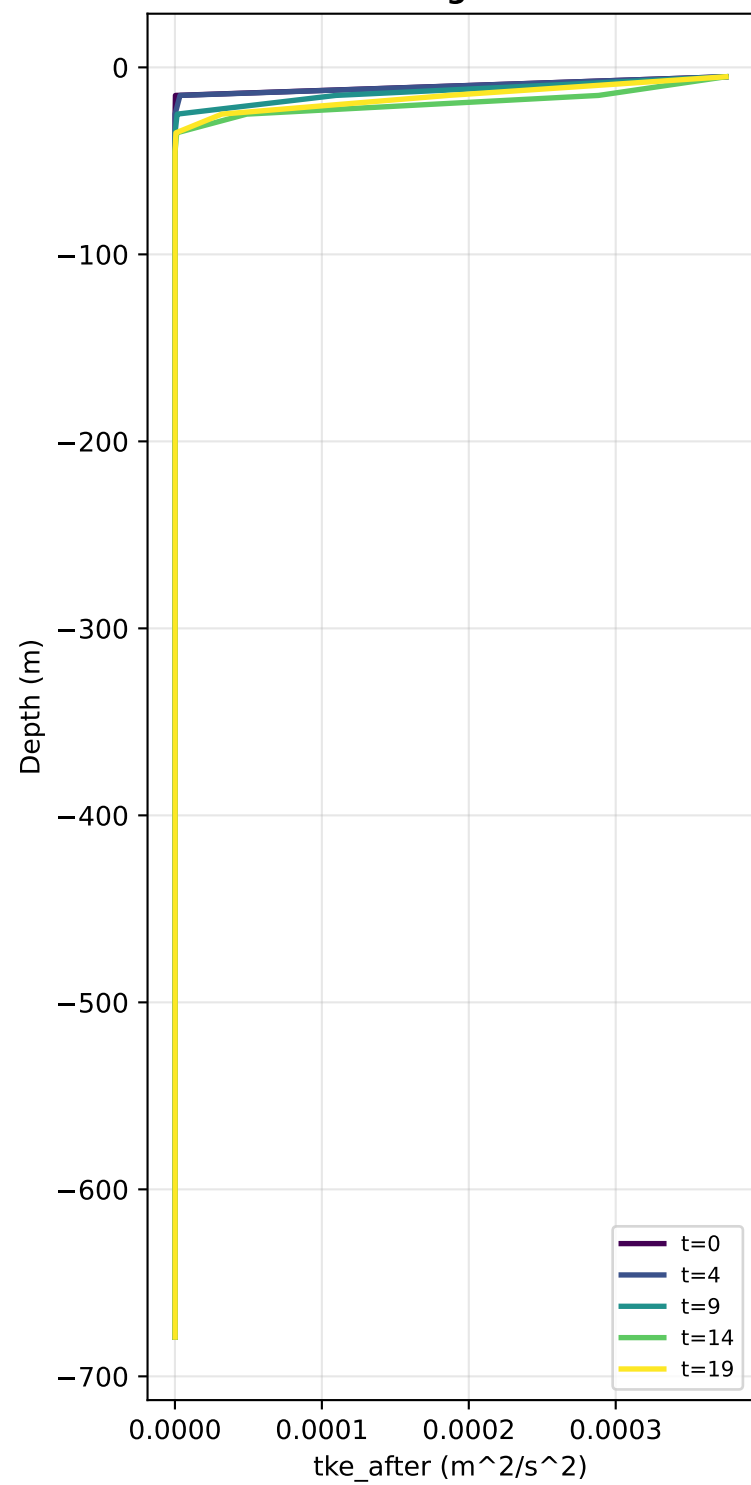
Median diff :	0.0000e+00
P95 diff :	1.0188e-09
Max diff :	7.8363e-07
Median rel:	0.0000e+00
P95 rel:	3.0206e-04
Max rel:	3.0660e-03
Fraction >1% rel:	0.0000%
Points compared:	520

Assessment:

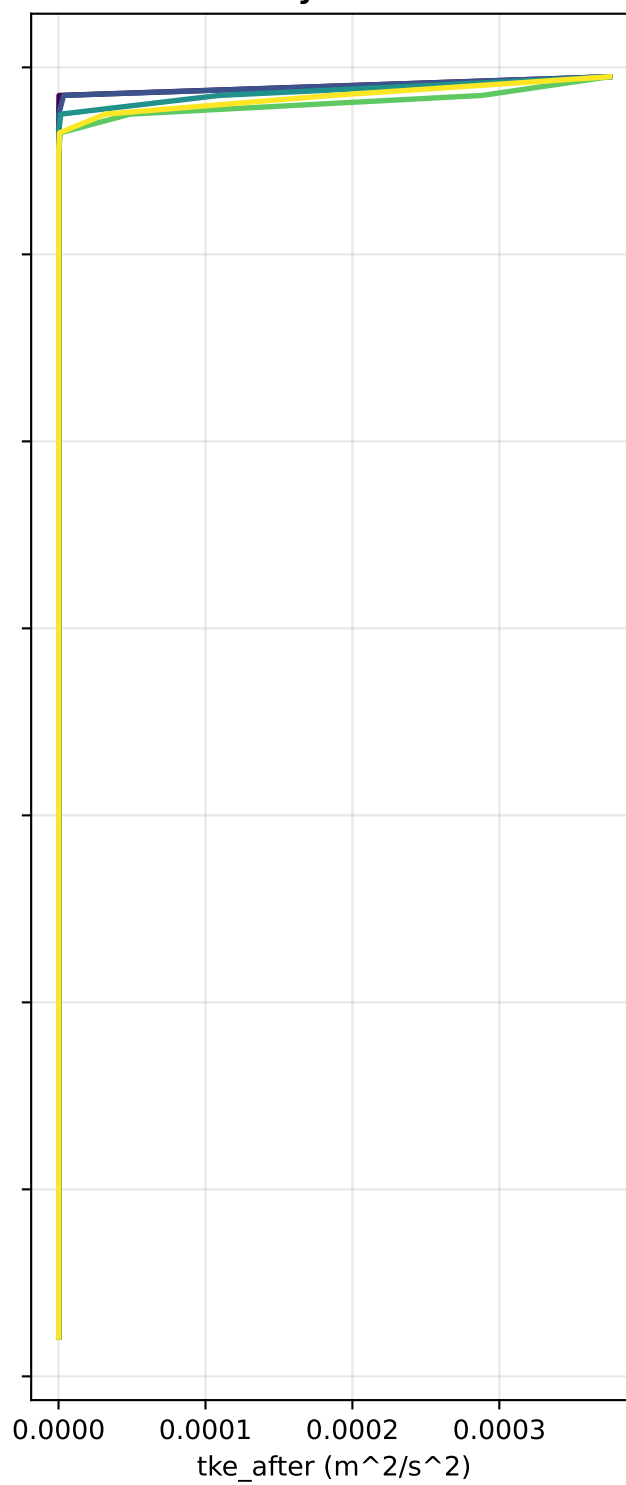
EXCELLENT

tke_after Vertical Profiles (TKE at t+1 (single-step replay))

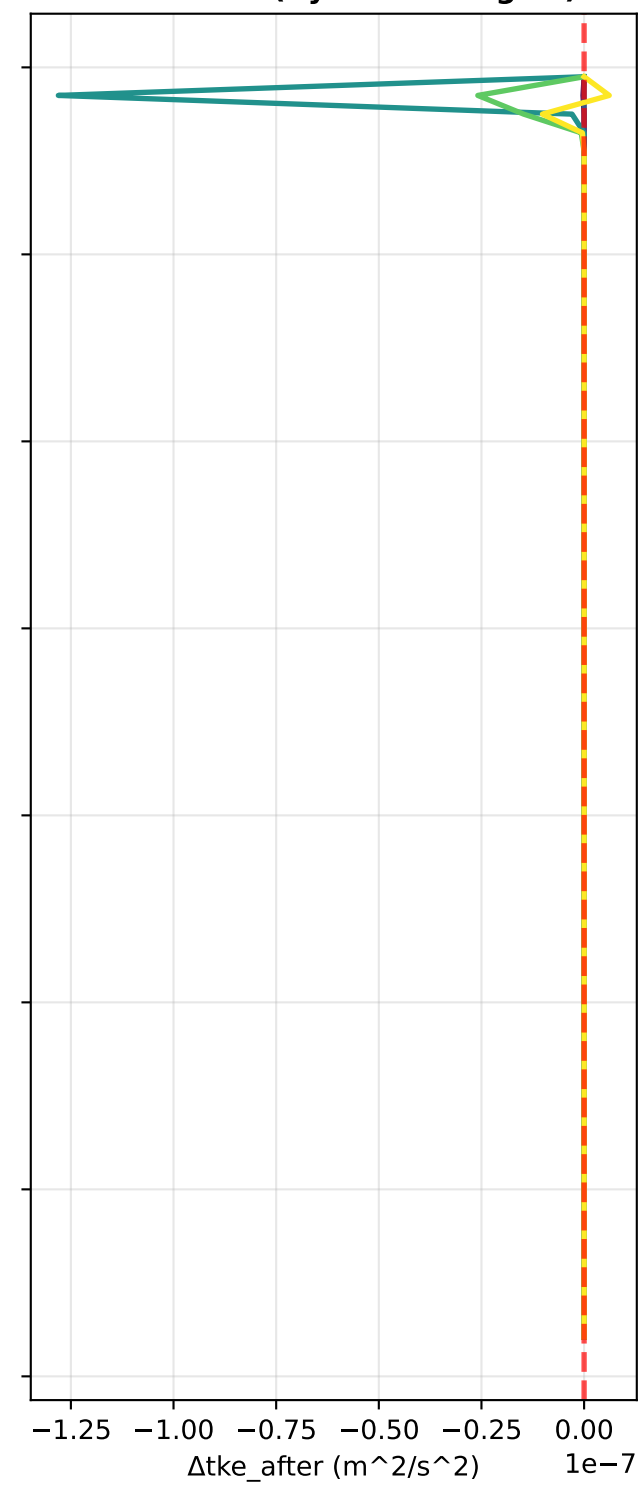
MITgcm



Python



Difference (Python - MITgcm)



Validation Summary & Assessment

OVERALL VALIDATION RESULT:

Status: ☒ PASS

COMPONENT RESULTS:

visc_az: ☒ EXCELLENT
Median rel error: 0.0000% Max |diff|: 1.3202e-05
Fraction >1% rel: 0.0000%

diff_kz: ☒ EXCELLENT
Median rel error: 0.0000% Max |diff|: 1.3202e-05
Fraction >1% rel: 0.0000%

mixing_length: ☒ EXCELLENT
Median rel error: 0.0694% Max |diff|: 7.7561e-03
Fraction >1% rel: 0.0000%

tke_after: ☒ EXCELLENT
Median rel error: 0.0000% Max |diff|: 7.8363e-07
Fraction >1% rel: 0.0000%

PROVENANCE:

UUID Match: ☒ Yes
MITgcm UUID: 84f5dfa8-4a53-4dae-9541-30dec502d26e
Python UUID: 84f5dfa8-4a53-4dae-9541-30dec502d26e

ACCEPTANCE CRITERIA (mirrors generate_kpp_validation_report.py's own):

☒ EXCELLENT: median rel error < 0.1%
☒ GOOD: median rel error < 1.0%
☒ CHECK: median rel error >= 1.0% -- may be a real, already-characterized gap (see closed_issues.md), not necessarily a new defect; this report states the number, not the cause.